CS341 · COMPUTER GRAPHICS

Mock Midterm Practice

20 marks · 1 hour · same format as the past midterms

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01 Questions

Q1 SHORT ANSWER · 4 PTS

1. What is a frame buffer?
2. Define persistence. Which kind of persistence is better for animation?
3. P = (4, 1) and Q = (1, 5). Find the vector from P to Q and its length.
4. A 10 × 6 inch screen has a resolution of 1280 × 720. Find the aspect ratio.

Q2 TRUE / FALSE · 4 PTS

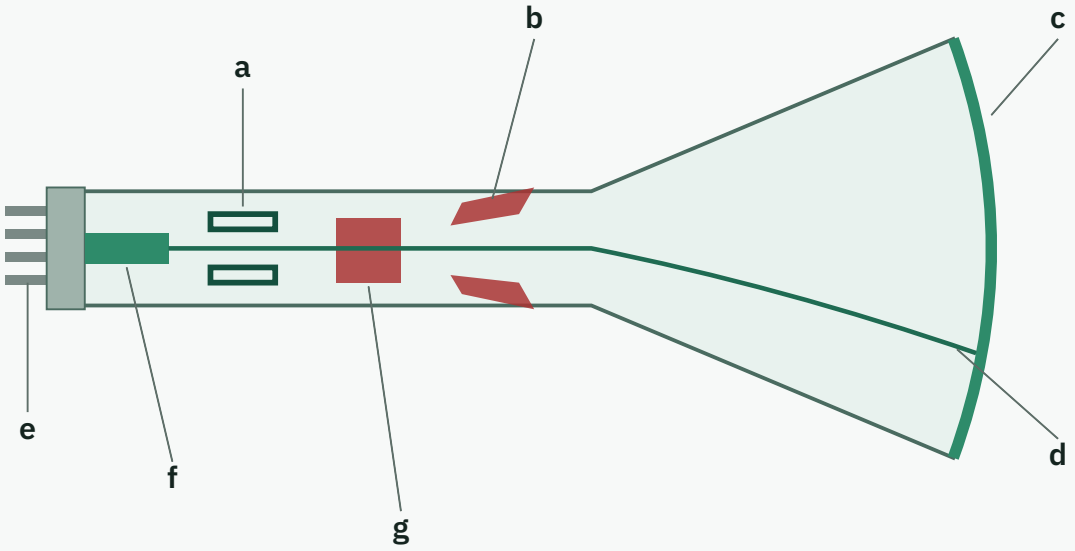
1. In a random scan system the electron beam directly draws the picture. ( )
2. A plasma panel is a nonemissive display. ( )
3. Point – Point = Vector. ( )
4. In the RGB cube, the origin (0, 0, 0) is white. ( )
5. Passive-matrix displays are cheaper than active-matrix displays. ( )
6. Bresenham's algorithm is more accurate than DDA because it avoids rounding. ( )
7. Tint is produced by adding black pigment to a pure color. ( )
8. The dot product of two vectors is a vector. ( )

Q3 MCQ · 4 PTS

1. Which device is a hard-copy device? a) Trackball b) Ink-jet printer c) Light pen d) Data glove
2. Magenta + Yellow (subtractive) = a) Green b) Blue c) Red d) Black
3. In the CIE chromaticity diagram, point C represents: a) Black b) White light c) Red d) The purple line
4. Two vectors are perpendicular when their dot product is: a) 1 b) –1 c) 0 d) Their length
5. The HSV model is: a) Hardware-oriented b) User-oriented c) Used for TV broadcasting d) Subtractive
6. glVertex2i(4, 5) specifies: a) 2 floats b) 2 integers c) 3 integers d) An integer array
7. Lines in raster systems look stair-stepped because: a) Low refresh rate b) Positions are rounded to pixels c) Phosphor decay d) Color mixing
8. The YIQ model is used in: a) Printers b) NTSC color TV c) LCD panels d) Pen plotters

Q4 CRT · 3 PTS

Label the cathode ray tube using the letters a–g:



Q5 BRESENHAM · 5 PTS

Using Bresenham's line algorithm, generate the pixels of the line from (1, 1) to (8, 5) and draw it. Show the steps.

k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				
6				

